

The real increment test

PISM is collapsing, and our ocean reacts violently to it

2026-07-13, evening. Follows *dynamics_seam_2026-07-13*.

Abstract

The previous note established that the CORE3(observed)→PISM cavity swap is a one-off monster that production never has to survive, and that cold-starting the ocean on PISM’s cavity sidesteps it. That was done, chunk 1 ran a clean model year, PISM ran its 10 years, and for the first time the remap was asked to carry *a year of genuine spun-up ocean dynamics across a real PISM increment*. It failed — but the way it failed inverts the conclusion. **The ocean absorbed ~99% of the change** (CFLz *peaked and then decayed*; η had zero NaN; only **17 of 209 172 nodes** misbehaved). What is actually wrong is upstream: **PISM is in massive disequilibrium and is shedding its ice shelves**. The Ross front retreats $\approx 4^\circ$ of latitude (≈ 440 km) in a *single* 10-yr leg, and **1399 nodes have their ice removed outright**, ringing the whole Antarctic margin. The “gentle 22m increment” framing used to justify this whole strategy was wrong. **This is not good news**: the run still crashed, and what it tells us is that our ocean reacts violently to ice-shelf and cavity changes mid-run. Absorbing 99% of the change is not a passing grade when the remaining 1% is fatal.

1 What the increment test actually did

Chunk-3 (awiesm3 1901) crashed at **day 15.66** (η_n NaN check). But read the diagnostics before concluding anything — they look nothing like the monster swap:

	monster swap (CORE3→PISM)	real increment (this leg)
CFLz over time	4.46 → 6.43 → 7.25 → 9.24 monotonic runaway	5.26 → 5.78 → 5.40 → 4.51 → 4.15 peaks, then DECAYS
died at	day ~4	day 15.7
η_n NaN	yes	none — all 209 172 nodes finite
$ \eta > 3$ nodes	3611, basin-wide, radiating mostly <i>unchanged</i> nodes	17 , one tight cluster all 17 on changed nodes

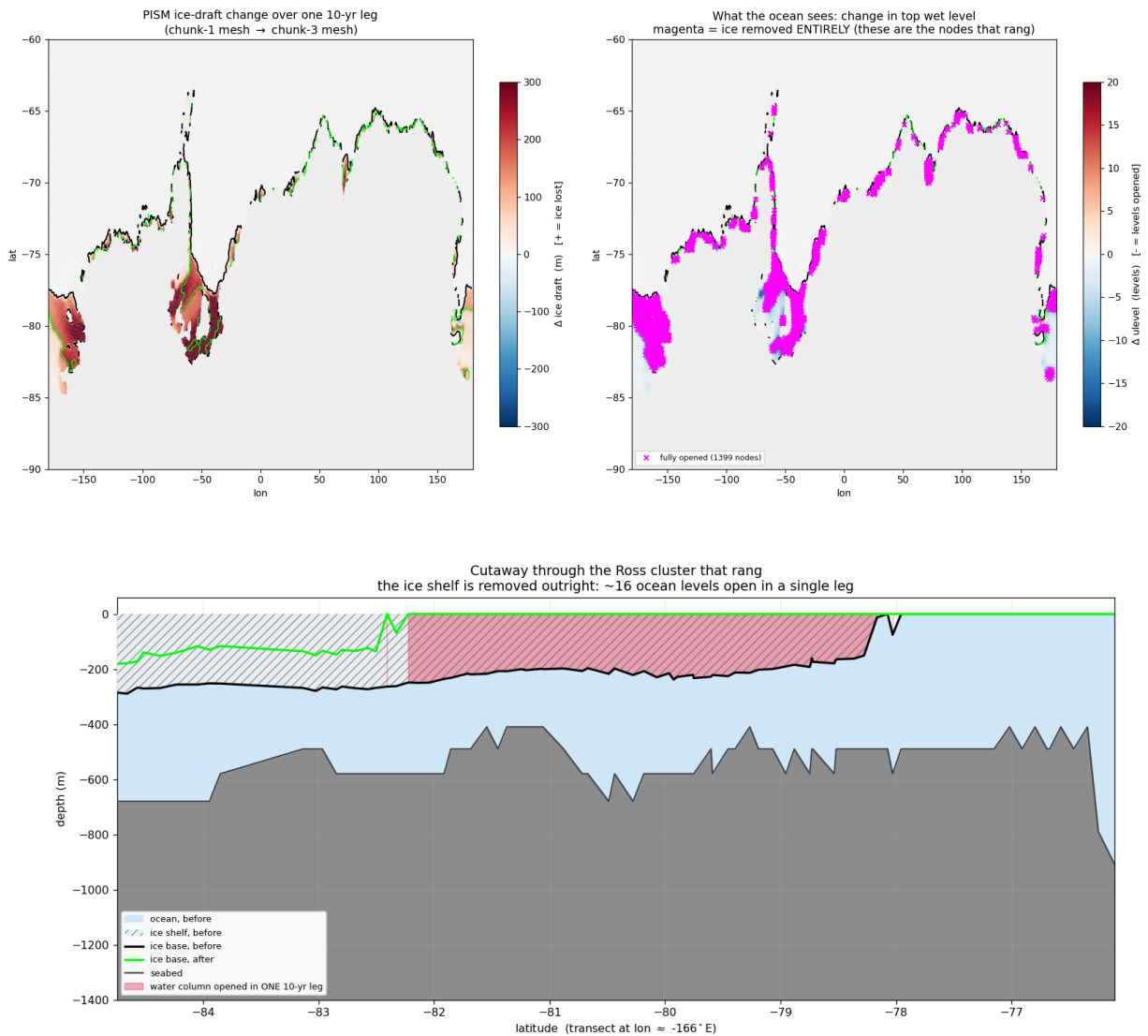
The CFLz *decays*: the ocean is *absorbing* the geometry change, not running away from it. And the 17 offending nodes share one signature exactly:

all 17 ringing nodes	ulev 17 → 1
i.e.	16 ocean levels opened <i>at once</i>
η values	+24.4, -17.9, +15.9, +8.3, +5.3
	alternating sign ⇒ grid-scale checkerboard

These are nodes where the ice shelf above them was **removed entirely** in one leg. They get donor water, zero velocity, and a free surface that cannot match their still-cavity neighbours ($\eta \equiv 0$ under ice) — and they ring. It is the same seam physics as before, but now confined to the *extreme tail* of the change rather than smeared over thousands of nodes.

2 The finding: PISM is collapsing

Why is the tail so extreme? Because the increment is not gentle at all.



The cutaway is the clearest statement of the problem: along this transect the ice base goes from $-200 \dots -280$ m to **zero (open ocean)** across the whole span -82.2° to -78.2° latitude. **The Ross ice-shelf front retreats $\approx 4^\circ$ of latitude (≈ 440 km) in ONE 10-yr leg.** The plan view shows this is not a Ross anomaly: the 1399 fully-opened nodes (magenta) ring the *entire* Antarctic margin.

changed nodes this leg	3248	(1.55% of mesh)
$ \Delta\text{ulev} \geq 10$	1582	
nodes with the ice removed ENTIRELY	1399	
mean $ \Delta\text{draft} $	2.9 m	$\leftarrow$ <i>meaningless</i>
max $ \Delta\text{draft} $	923 m	
PISM floating cells	24 815 $\rightarrow$ 20 093	(-24% in 10 yr)

The mean is a lie. It is diluted by the vast unchanged interior; the distribution is wildly skewed. The earlier claim that a 10-yr PISM increment is a gentle ≈ 22 m change — which is what motivated

the entire “sidestep the swap and the increments will be fine” strategy — was **wrong**, and wrong in a way that mattered.

PISM, started from the inherited spin-up state, is **not in equilibrium**: it is shedding ice shelves catastrophically. No ocean model should be expected to swallow 440 km of ice-front retreat in one coupling step.

3 This is not good news

It is tempting to read the numbers optimistically: of the **1399** nodes whose ice was removed outright, only **17 rang**; the ocean absorbed $\sim 99\%$ of even *this*; the vertical-CFL transient peaked and *decayed*. All true, and it does mean the remap machinery — bit-exact on unchanged nodes, inversion-free, geometrically sound, and (after §4) free of manufactured values — is performing better than a “crashed at day 15” headline suggests.

But the honest reading is negative. The run still *crashed*. It got further, and it crashed. What we have learned is that **our ocean reacts violently to ice-shelf and cavity changes mid-run**: it does not need many bad nodes to kill a leg, and a handful of columns where the ice vanishes outright is enough. A coupled setup in which the ice sheet is *allowed* to change its cavity every 10 years cannot be one leg’s worth of extreme nodes away from failure. Absorbing 99% of the change is not a passing grade when the remaining 1% is fatal.

4 A real remap bug fixed on the way

The first attempt at this leg died at **mstep 1** with found `temperature becomes NaN or <-5.0, >60`. The remapped restart contained -45.02°C — a value that exists *nowhere* in the source (wet range $[-2.32, 41.72]$, dry cells all 0.0). The remap *manufactured* it. The column made the mechanism obvious:

1.22, -0.99, -3.19, -5.39, -8.69, -13.09, -18.6, -25.2, -34.01, **-45.02**

Smooth and accelerating: an **unbounded linear extrapolation**. For columns that get *deeper* across a mesh change, the remap fitted a straight line through the last two old levels and continued it downward with no clamp; over ~ 9 new levels a thermocline slope compounds into nonsense. Fixed by taking such water from the *nearest old node actually wet at that depth* — the horizontal donor the rest of the remap already uses. Water below the old seabed should look like the nearby ocean at that depth, not like a continuation of the local gradient.

Never hit before because every previous remap went mother $\rightarrow$ submesh: the new mesh was a strict *subset*, so no column ever got deeper. Ice *retreating* (208 810 $\rightarrow$ 209 172 nodes) exercises this path for the first time. (esm_tools eef4f6db.)

5 Where this leaves the strategy

Two branches of work, plus the thing that would actually fix it for good.

1. **Find out why PISM is collapsing, and stop it.** A run shedding 24% of its floating area in a decade is the anomaly, and everything downstream inherits it. **The first thing to check is the sub-shelf melt we feed PISM**: if our ocean is handing it unrealistic melt, then PISM’s collapse is our own doing, and the loop closes. Until PISM is behaving, the ocean is being asked an unreasonable question every leg.

2. **Try to stabilise the ocean locally.** Reduce the ocean timestep and/or raise the viscosity *in the neighbourhood of the changed columns* and see whether the transient can be damped enough to survive. This is a mitigation, not a cure — but if a handful of extreme nodes is all that stands between us and a completed leg, it may be enough to keep the coupling running while (1) is sorted out. Note the global knobs are already exhausted: a global timestep cut does *not* help (60s merely swaps the CFLz symptom for an η_n one).
3. **The lasting way forward: compute physically consistent fields for the newly opened cavity.** The 17 failures all have the same shape — a column where the ice vanishes, handed donor water, zero velocity, and a free surface that cannot match its still-cavity neighbours ($\eta \equiv 0$ under ice). Patching such a column with plausible *values* is what we have been doing, and it is exactly what does not work (five fill strategies, all dead; see `dynamics_seam_2026-07-13`). What is needed is a state for the newly opened water that is *dynamically consistent with its surroundings* — not merely a sensible number in each cell. That is the real problem, and it is the one worth solving.

A per-leg cap on $|\Delta_{\text{ulev}}|$ (defer the extreme tail across legs) remains available as a blunt robustness guard, but it papers over (1) rather than fixing anything.

6 Method note (continued)

The “mean $|\Delta_{\text{draft}}| = 22\text{ m}$ ” number was computed correctly and reported honestly — and was still deeply misleading, because a mean over a domain that is 98% unchanged says nothing about the tail that actually breaks the model. The distribution should have been plotted before it was used to justify a strategy. Two figures did in ten minutes what a summary statistic obscured for a day.